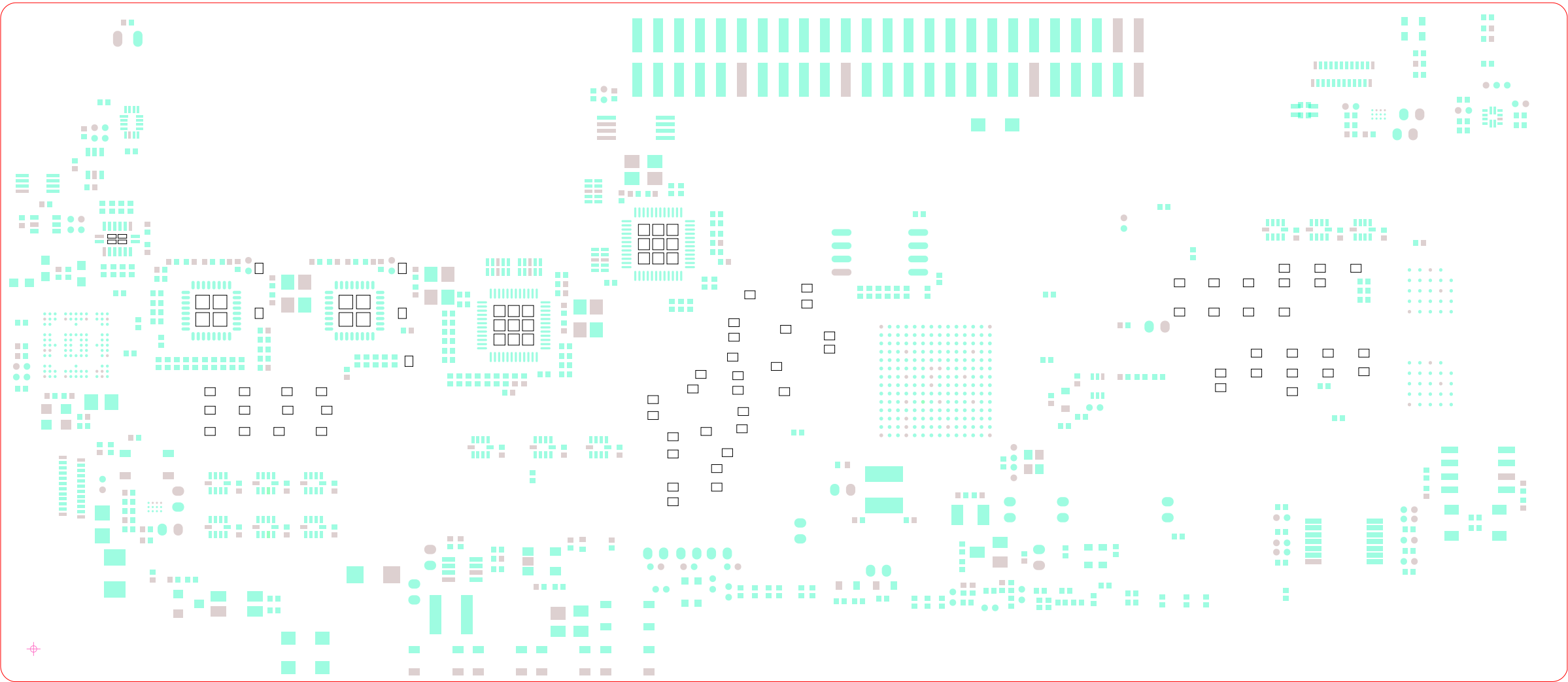
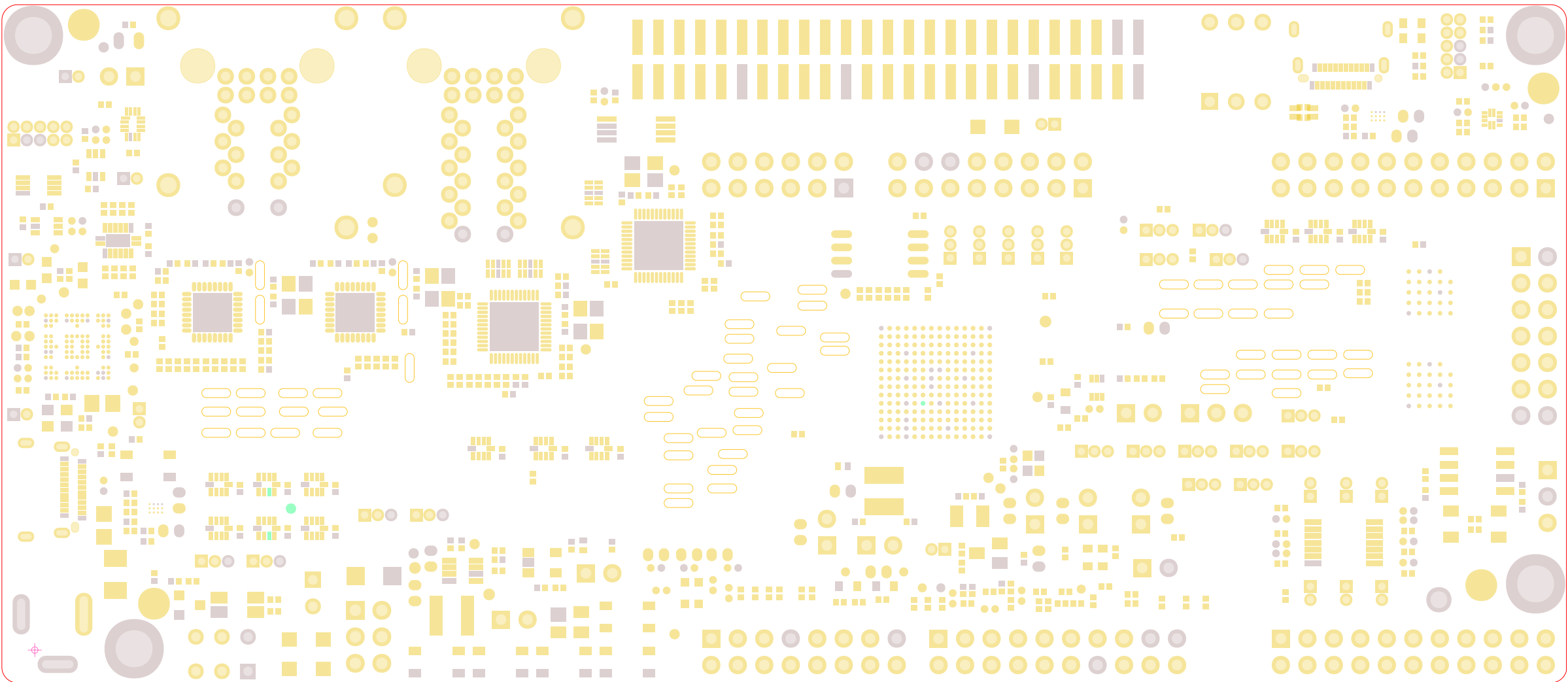
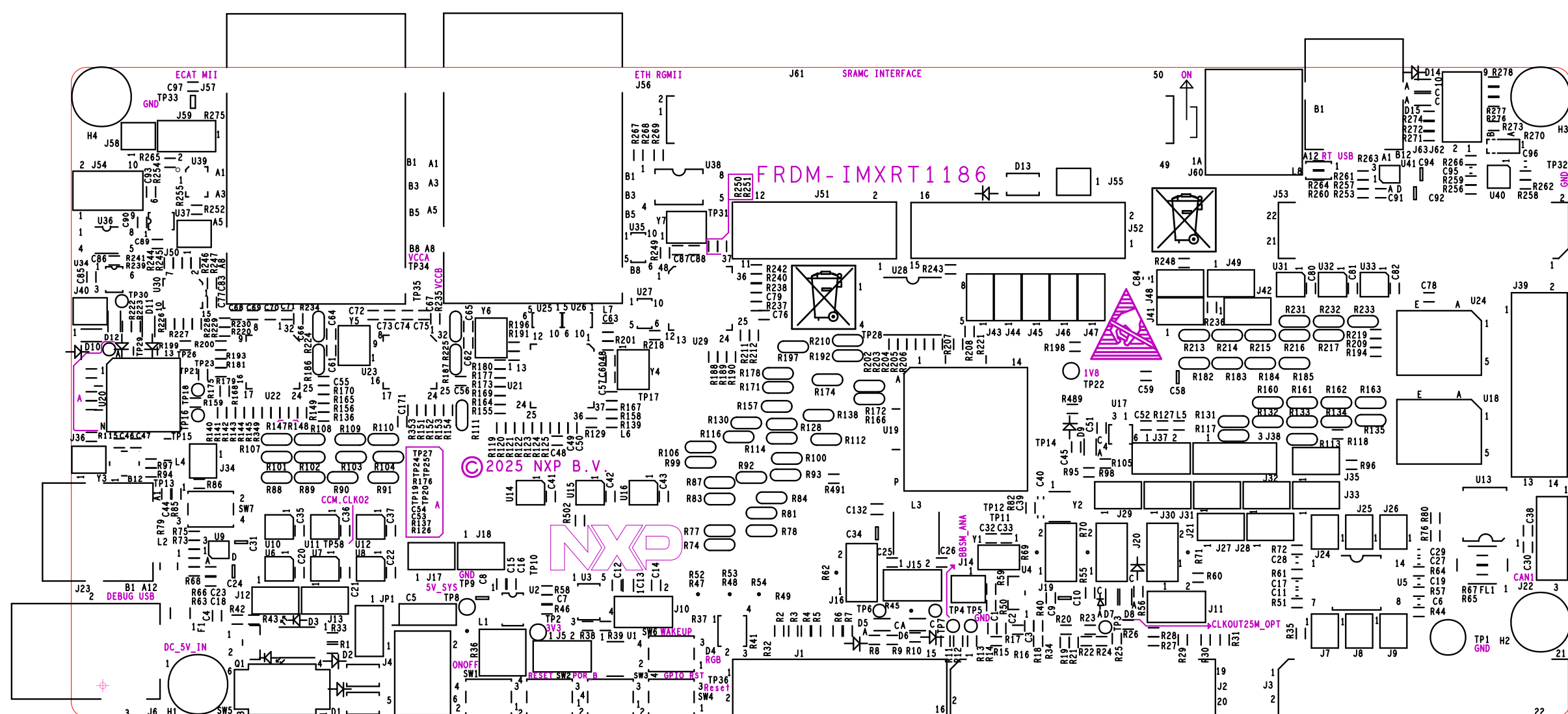




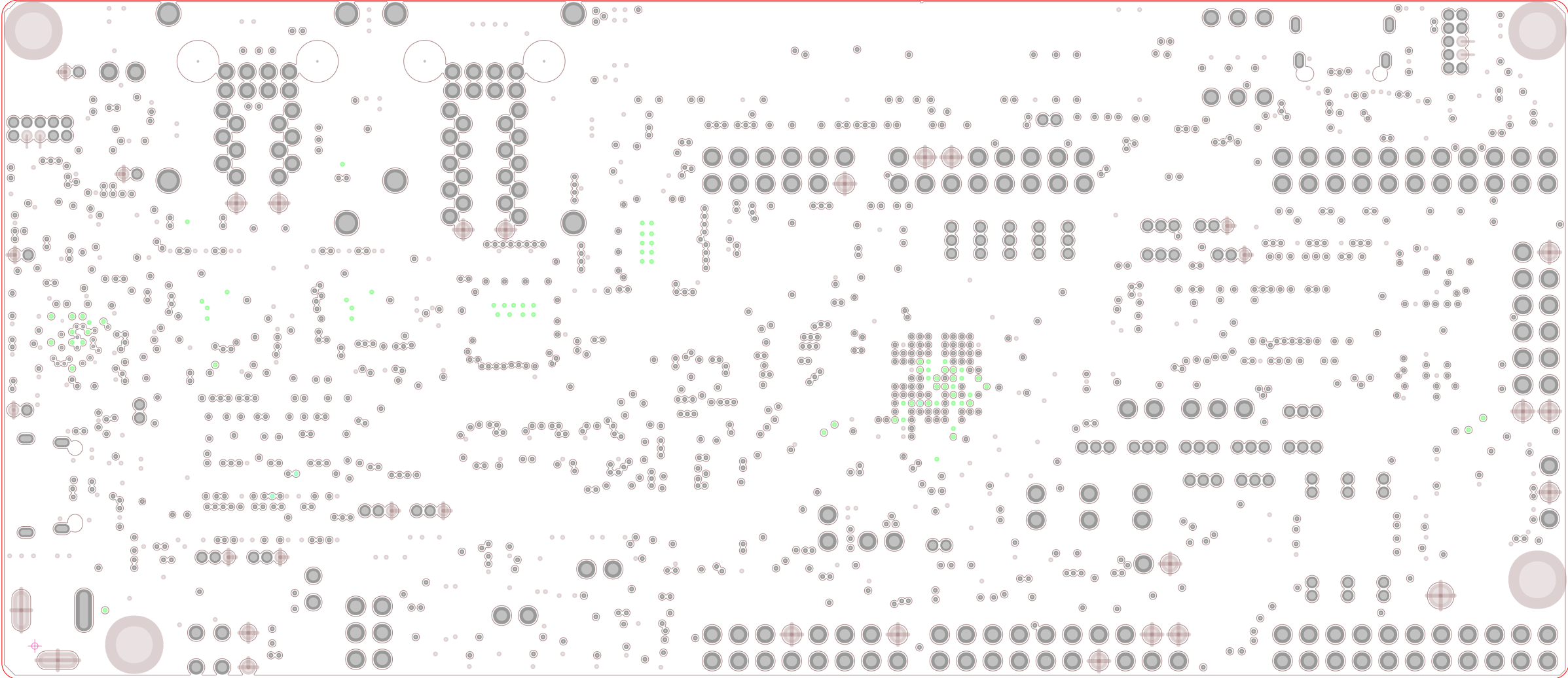
NXP SEMICONDUCTOR 6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA		NAME FRDM-IMXRT1186		
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		☐ FCP (NXP CONFIDENTIAL PROPRIETARY)		
SHEET 1 OF 12	FILM PRIMARY PASTEMASK	DATE 09-05-25	NUMBER 170-95302	REV A2



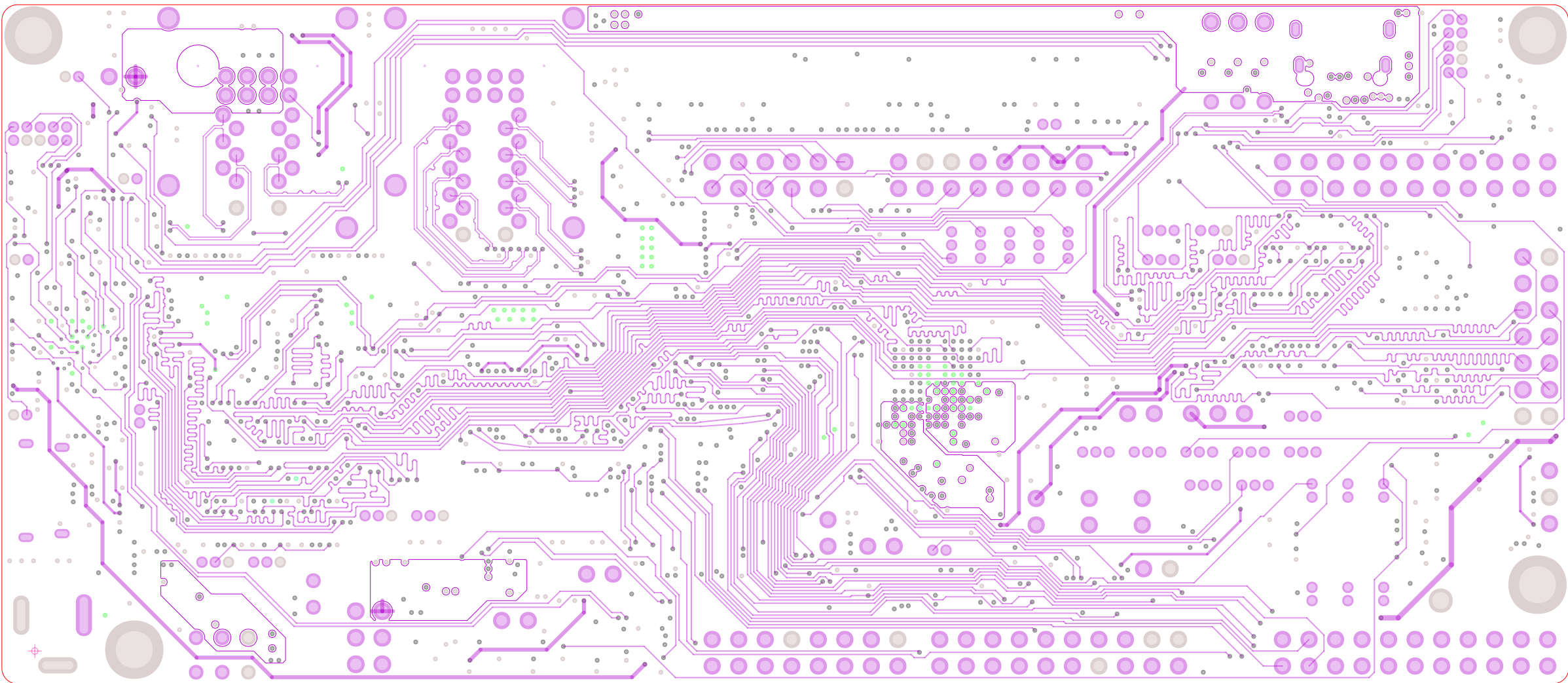
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SHEET 3 OF 12	FILM PRIMARY SOLDMASK	DATE 09-05-25	NUMBER 170-95302 REV A2



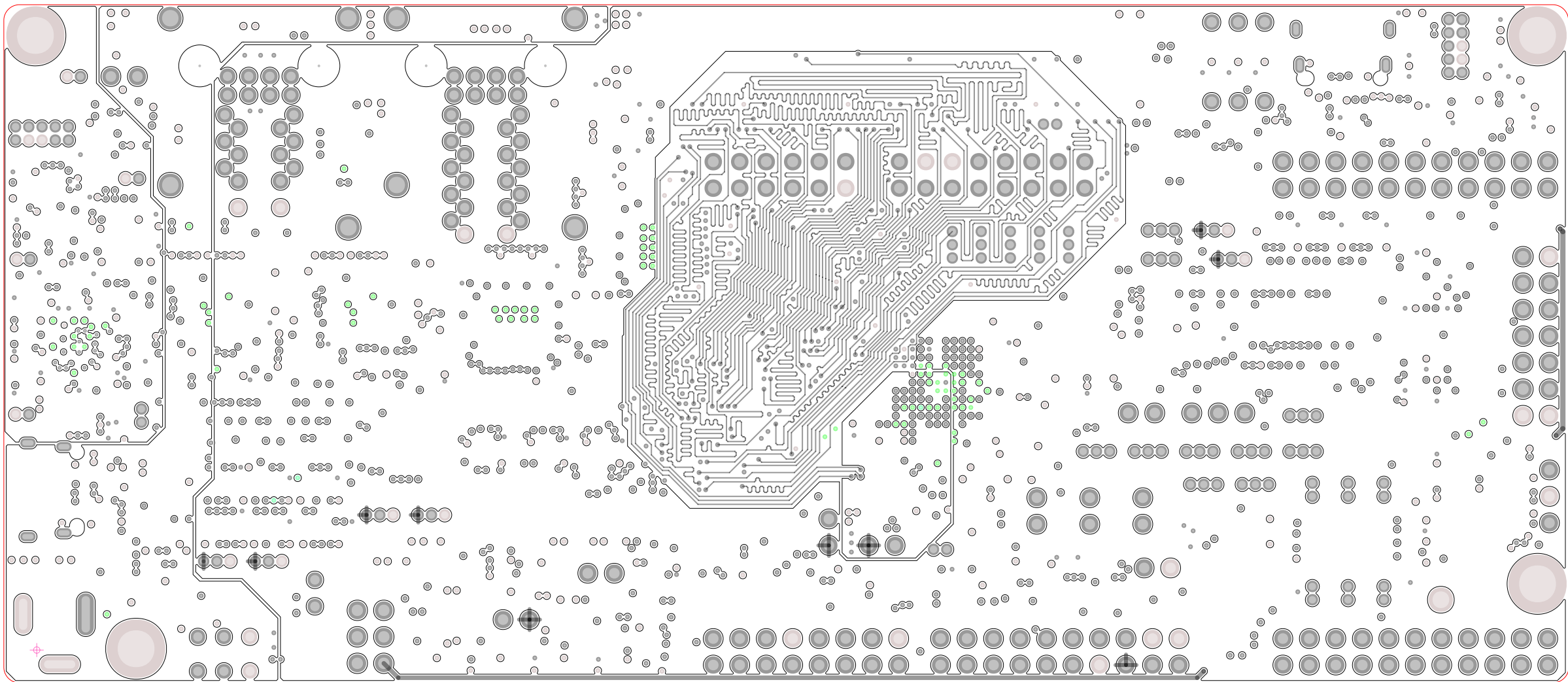
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SHEET 2 OF 12	FILM PRIMARY SILKSCREEN	DATE 09-05-25	NUMBER 170-95302 REV A2



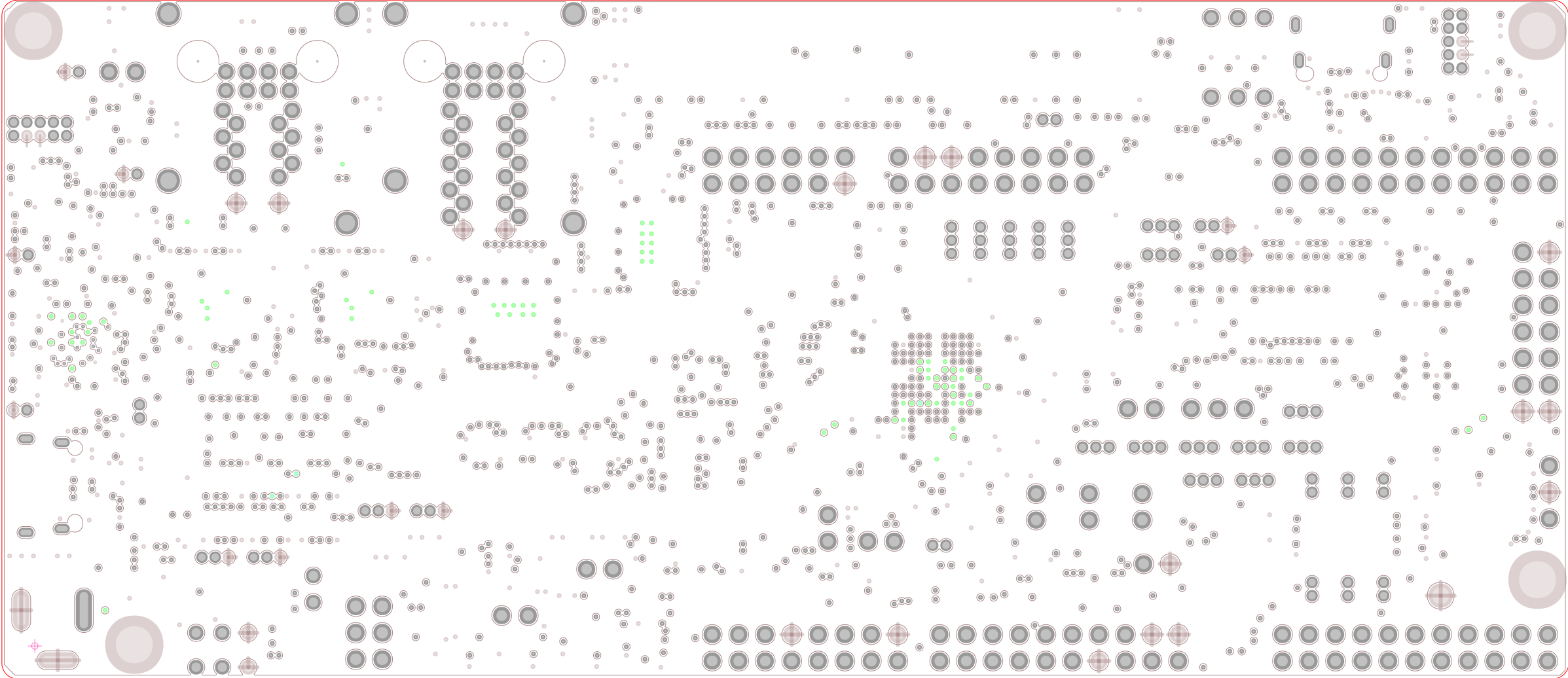
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SHEET 5 OF 12	FILM L2_GND_1	DATE 09-05-25	NUMBER 170-95302 REV A2



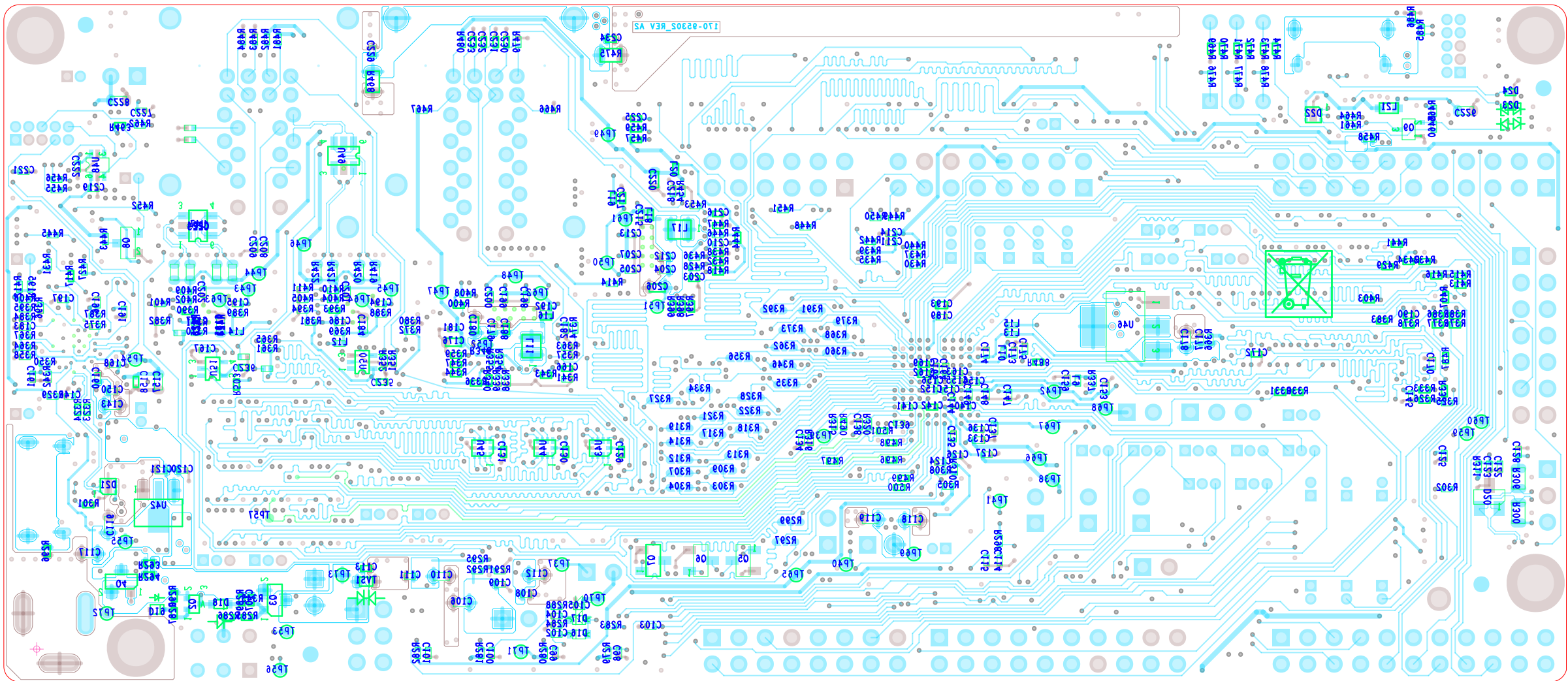
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SHEET 6 OF 12	FILM L3_INT_1	DATE 09-05-25	NUMBER 170-95302	REV A2



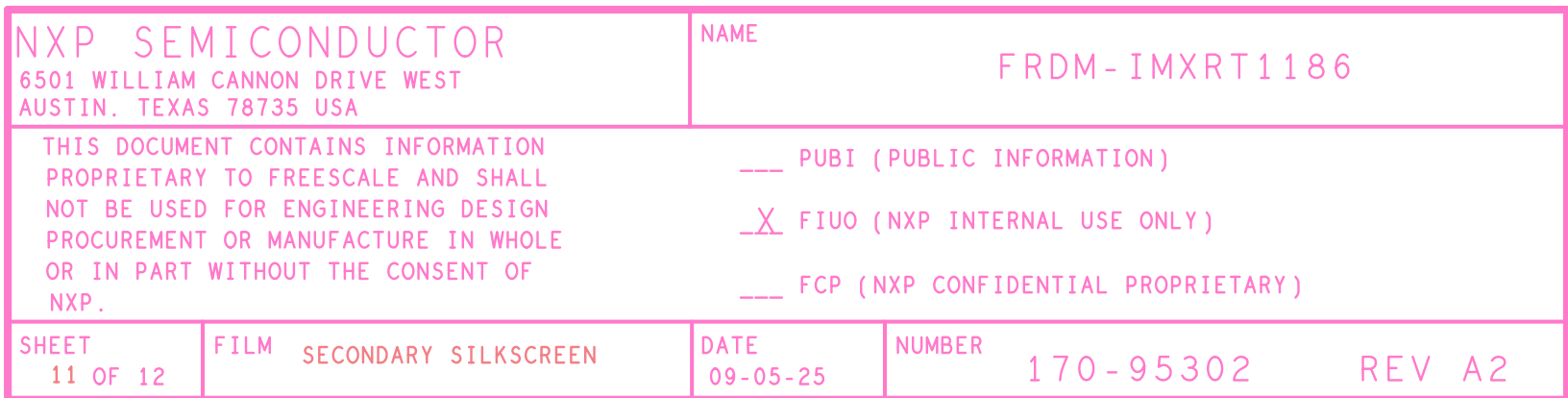
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SHEET OF 12	FILM	DATE 09-05-25	NUMBER 170-95302	REV A2

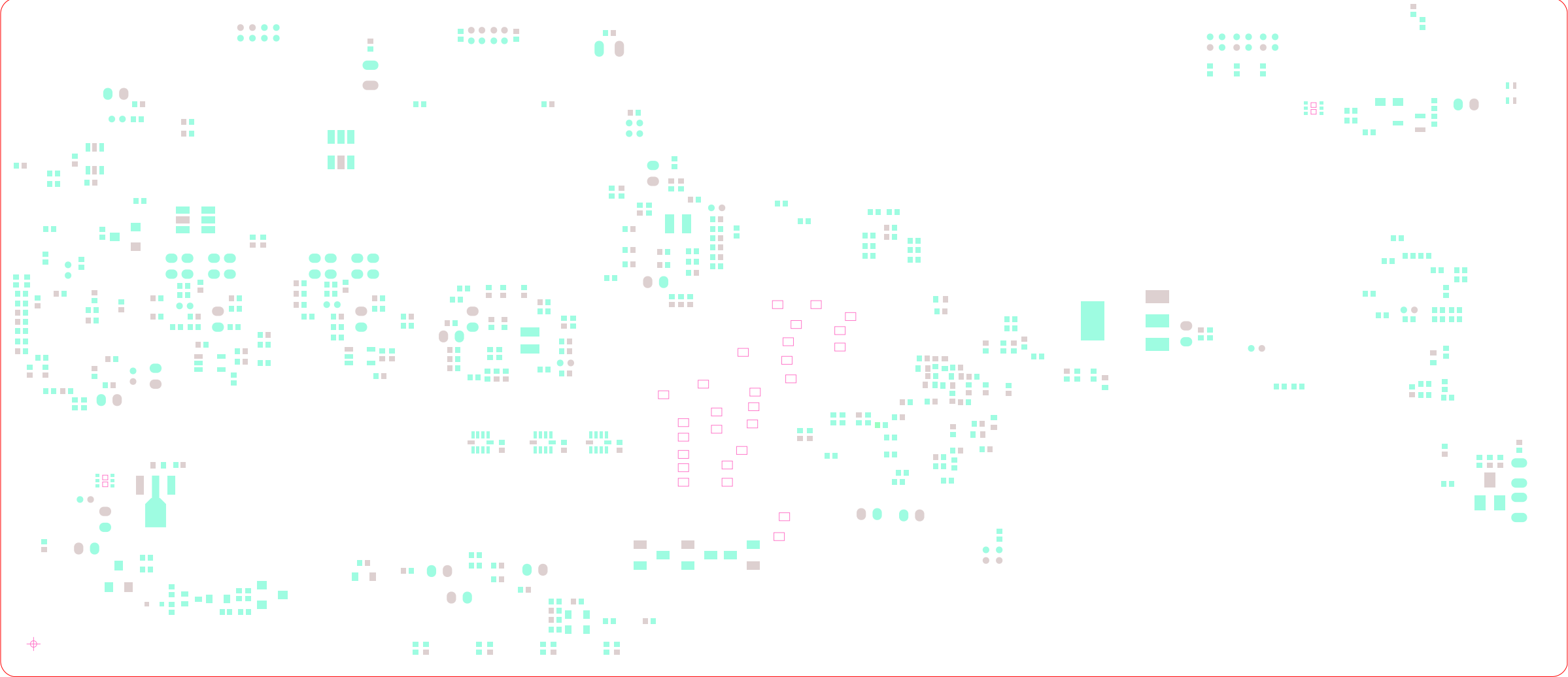


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SHEET OF 12	FILM	DATE 09-05-25	NUMBER 170-95302 REV A2

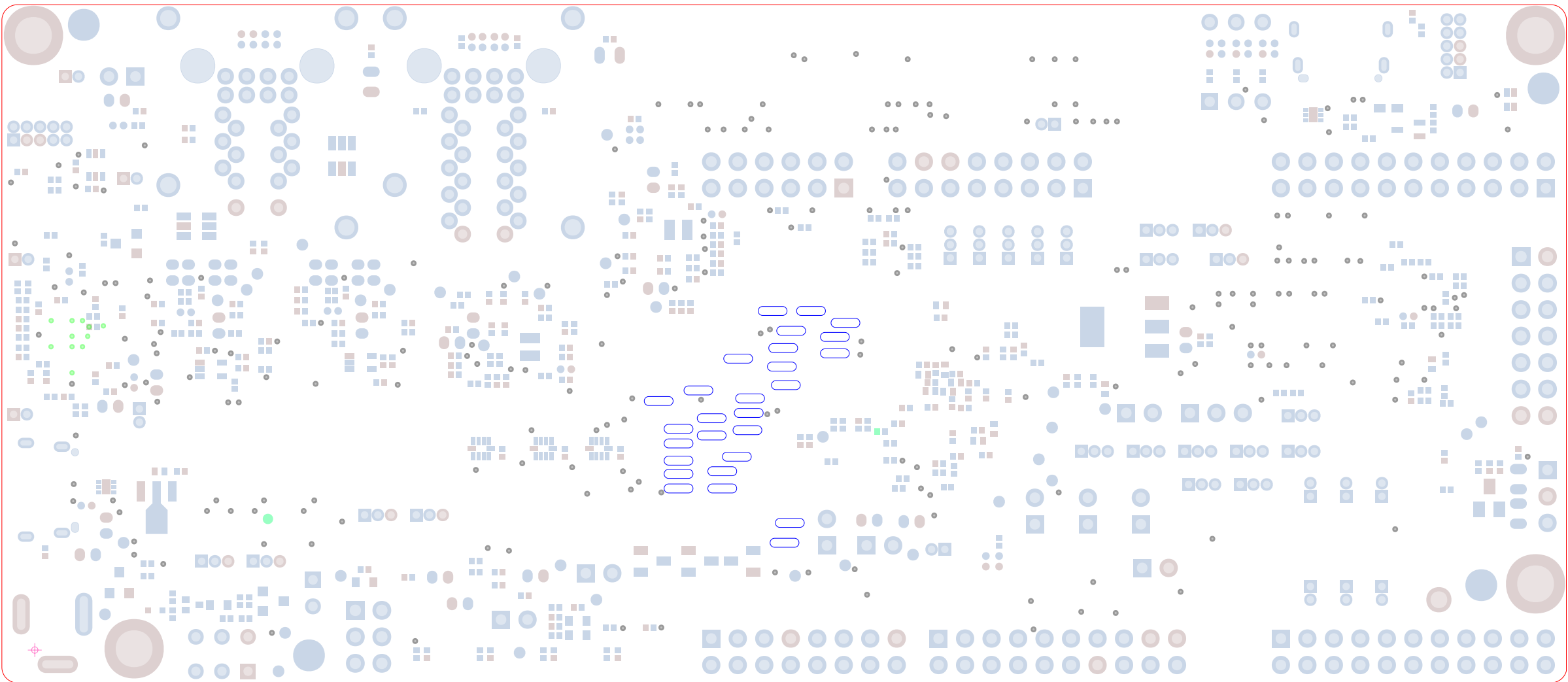


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SHEET 9 OF 12	FILM L6_SS	DATE 09-05-25	NUMBER 170-95302	REV A2





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SHEET 12 OF 12	FILM SECONDARY PASTEMASK	DATE 09-05-25	NUMBER 170-95302 REV A2



NXP SEMICONDUCTOR 6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA			NAME FRDM-IMXRT1186	
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SHEET 10 OF 12	FILM SECONDARY SOLDMASK	DATE 09-05-25	NUMBER 170-95302	REV A2

		2		1		
		REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED		
				DE	PE	CAD
	A	ORIGINAL RELEASE	03-14-25	A.L.	L.P.	A.L.
	A1	LAYOUT UPDATED VENDOR COMMENTS	03-20-25	A.L.	L.P.	A.L.
	A2	DESIGN UPDATED REWORK	09-05-25	A.L.	L.P.	A.L.

THIS DRAWING SPECIFIES THE REQUIREMENTS FOR A PRINTED WIRING BOARD IN ACCORDANCE WITH SPECIFICATION IPC-6012 CLASS 2 (LATEST REVISION).

- THE PWB MUST BE LEAD FREE ASSEMBLY PROCESS COMPATIBLE AND MUST BE ABLE TO HANDLE A MINIMUM OF 5 CYCLES AT 260 DEGREES CELSIUS FOR 10 SECONDS.
- BASE MATERIAL - LAMINATE AND PREPREG SHALL MEET IPC-4101D-26, 83 or 98
T_g - MUST BE GREATER THAN OR EQUAL TO 150 DEGREES CELSIUS.
T_d - MUST BE GREATER THAN OR EQUAL TO 330 DEGREES CELSIUS.
- COPPER FOIL WEIGHT - SEE STACKUP DETAIL 'A'
- CHARACTERISTIC IMPEDANCE - SEE DETAIL 'B'
- MINIMUM CONDUCTIVE WIDTH/SPACING TO BE .0032"/.004"
- PLATING FINISH: A. BOTH SIDES ENIG; TO MEET THE REQUIREMENTS OF IPC-4552 (LATEST REVISION).
- FAB VENDOR IS NOT ALLOWED TO USE ODB FOR FABRICATION. CAN BE USED ONLY FOR REFERENCE.
- SOLDERMASK - TO MEET THE REQUIREMENTS OF IPC-SM-840E (OR LATEST REVISION).
BLACK COLOR, BOTH SIDES. MODIFICATION OF SOLDERMASK IS NOT ALLOWED WITHOUT WRITTEN PERMISSION FROM NXP.
TYPE: LPI OR EQUIVALENT.
A. LOCATION = +/- .002" OF PLATED PADS.
B. DIAMETER OR SIZE = +/- .002 OF ORIGINAL DATA

10. SILKSCREEN - WHITE EPOXY OR ACRYLIC INK, BOTH SIDES, NO SILKSCREEN ON ANY EXPOSED COPPER FEATURE.
11. ELECTRICAL TEST - 100% IPCD356, PCB FABRICATOR TO PERFORM A NET COMPARE AGAINST THE IPCD356 NETLIST PROVIDED BY NXP.
12. PRINTED WIRING BOARD IS TO BE INDIVIDUALLY BAGGED.
13. DFM CHECK MUST BE RUN ON BOARD DATA BEFORE BUILDING BOARDS,
UNLESS PRIOR APPROVAL IS GIVEN IN WRITING BY NXP.
14. TEARDROPS MAY BE ADDED AT THE FAB HOUSE TO ALL SIGNAL LAYERS.
15. TWO SOLDER SAMPLES TO BE PROVIDED.

16. SUPPLIER MARKINGS - ON SECONDARY SIDE ONLY, WHERE SHOWN.
- MUST BE UL RECOGNIZED AND MUST HAVE AN ID THAT CONFORMS TO UL94V-0

17. THE PWB WILL BE MARKED AS LEAD FREE BY USE OF AN INK STAMP ☒ **P6**

18. THE PWB WILL BE MARKED AS LEAD FREE PROCESS COMPATIBLE BY USE OF AN INK STAMP (260°C)

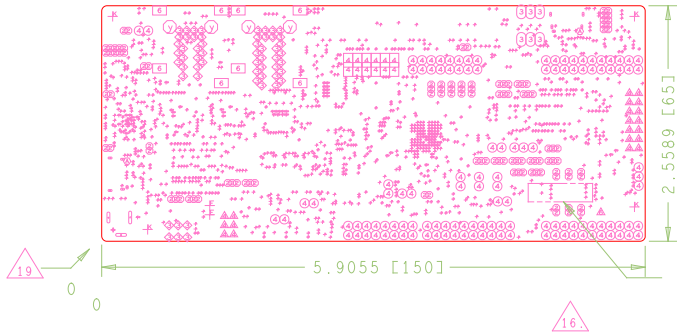
19) ALL PLATED AND NON-PLATED THROUGH HOLES ARE TO BE DRILLED AT PRIMARY DRILL STEP.
ALL HOLE LOCATION TOLERANCES ARE TO BE $\pm .002$ IN REFERENCE TO THE PRIMARY DATUM
UNLESS OTHERWISE SPECIFIED.

20. FINISHED PCB MUST BE PANELIZED FOR ASSEMBLY ACCORDING TO CONTRACT MANUFACTURERS REQUIREMENTS. THE ADDITION OF RAILS AND .125" NON-PLATED TOOLING HOLES ARE AT THE DISCRETION OF CONTRACT MANUFACTURER. PANELIZATION MUST BE APPROVED BY CONTRACT MANUFACTURER.

21. THE MANUFACTURE HAS THE OPTION TO ADD COPPER THIEVING ON OUTER AND INNER LAYERS, KEEP A MINIMUM DISTANCE OF .100" FROM ANY BOARD FEATURES.

22 THIS BOARD USES VIA-IN-PAD: SEE FAB_VIAFILL.ART

- A. ALL VIAS USING 8.1 DRILL SIZES ARE TO BE FILLED WITH NON-CONDUCTIVE VIA FILL. LACKWERKE-PETERS PP2795 OR EQUIVALENT AND MADE PLANAR TO THE PADS.
- B. OVERPLATE THE FILLED VIA AND APPLY FINISH METAL TREATMENT.
- C. DIMPLE ON VIA-IN-PADS MUST BE NO GREATER THAN .003" AND PROTRUSION NO GREATER THAN .002"



PART NO. 170-95302		NXP SEMICONDUCTORS			
--- COMPANY PUBLIC ☒ COMPANY INTERNAL --- COMPANY CONFIDENTIAL		THIS DOCUMENT CONTAINS INFORMATION PROPRIETARY TO NXP AND SHALL NOT BE USED FOR ENGINEERING DESIGN PROCUREMENT OR MANUFACTURE IN WHOLE OR IN PART WITHOUT THE CONSENT OF NXP.		6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA	
ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL NOT CONTAIN ANY OF THE DIRECTIVE 2011/65/EU (EU) ANNEX II TO DIRECTIVE 2011/65/EU: A REQUIRED UPON REQUEST.		UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ARE: ANGLES DECIMALS J0-J30" JXX .01 JXXX .005 ☒ ALL MACHINED SURFACES BREAK/ ALL SHARP EDGES AND CORNERS. REMOVE BURRS UNDERLINED DIM. NOT TO SCALE. THIRD ANGLE ORTHOGRAPHIC PROJECTION IS USED.		TITLE: PRINTED WIRING BOARD FRDM-IMXRT1186	
ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL NOT CONTAIN ANY OF THE DIRECTIVE 2011/65/EU (EU) ANNEX II TO DIRECTIVE 2011/65/EU: A REQUIRED UPON REQUEST.		APPROVALS DATE DRAWN ALBERT LI 09-05-25 CHECKED LEO PAN 09-05-25 DESIGN ENGINEER ALBERT LI 09-05-25		SIZE CAD FILE NAME DWG. NO. REV D LAY-95302 FAB-95302 A	
		SCALE 1 / 1 DO NOT SCALE DRAWING		SHEET 1 OF 2	

8	7	6	5	4	3	2	1			
NOTES (UNLESS OTHERWISE SPECIFIED):				REVISIONS						
				ZONE	REV	DESCRIPTION	DATE	APPROVED		
								DE	PE	CAD
	A	ORIGINAL RELEASE		03-14-25	A.L.	L.P.	A.L.			
	A1	LAYOUT UPDATED VENDOR COMMENTS		03-20-25	A.L.	L.P.	A.L.			
	A2	DESIGN UPDATED REWORK		09-05-25	A.L.	L.P.	A.L.			
THIS FAB_VIAFILL.ART SHOWS LOCATIONS OF VIA_IN_PAD TO BE FILLED.										
ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL MEET THE RFP COMMITTEE/COMMISSION DELEGATED DIRECTIVE (CDU) 2015/443 OF 31 MARCH 2015 AMENDING ANNEX I (A TO DIRECTIVE 2011/65/EU). A REQUESTED UPON REQUEST.				UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ARE: DECIMALS ANGLES .XX .01 .0-90° .XXX .005 ✓ RMS ALL MACHINED SURFACES BREAK ALL SHARP EDGES AND CORNERS. REMOVE BURRS. UNDERLINED DIM. NOT TO SCALE. THIRD ANGLE ORTHOGRAPHIC PROJECTION IS USED.						
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COMPANY PUBLIC X COMPANY INTERNAL COMPANY CONFIDENTIAL				APPROVALS DATE DRAWN ALBERT LI 09-05-25 CHECKED LEO PAN 09-05-25 DESIGN ENGINEER ALBERT LI 09-05-25						
TITLE 6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA PRINTED WIRING BOARD FRDM-IMXRT1186				SIZE CAD FILE NAME DWG. NO. REV D LAY-95302 FAB-95302 A						
SCALE 1/1 DO NOT SCALE DRAWING SHEET 2 OF 2										